

Module 08: Planning — Sophisticated Inference, Deep Temporal Models, and Hierarchical POMDPs

Learning Objectives

1. Derive **sophisticated inference**: recursive belief updating where the agent models its own future beliefs and actions.
2. Formalize **deep temporal models** (multi-step planning) through policy trees and Expected Free Energy evaluation.
3. Introduce **hierarchical POMDPs** for multi-timescale planning.

Introduction

Planning is the most computationally demanding component of Active Inference — it requires the agent to evaluate the consequences of action sequences extending into the future. This module develops three levels of mathematical sophistication: basic policy evaluation (flat EFE), sophisticated inference (recursive belief updating), and hierarchical POMDPs (multi-timescale planning).

Key Concepts

1. Deep Temporal Planning

The Expected Free Energy for a policy π of length T is:

$$G(\pi) = \sum_{\tau=t}^T G_{\tau}(\pi)$$

where each term evaluates the expected free energy at future time τ under policy π . This requires:

1. **State prediction**: $q(s_{\tau} | \pi) = \sum \{s | \pi\} B(\pi_{\tau}) \cdot q(s_{\tau-1} | \pi)$
2. **Outcome prediction**: $q(o_{\tau} | \pi) = A \cdot q(s_{\tau} | \pi)$
3. **EFE evaluation**: $G_{\tau}(\pi) = E_{\pi}[\ln q(s_{\tau} | \pi) - \ln p(o_{\tau}, s_{\tau})]$

The number of policies grows exponentially with planning horizon: K^T policies for K actions over T steps. This necessitates pruning strategies.

2. Sophisticated Inference

Sophisticated inference (Friston et al., 2021) extends basic policy evaluation by having the agent model *what it will believe* in the future:

$$q(s_{t+1} | \pi) \rightarrow q(s_{t+1} | o_{t+1}, \pi)$$

The agent simulates not just future states but future observations and the belief updates they will produce. At each future timestep:

1. Predict future observation: $q(o_{t+1} | s_{t+1})$
2. Update future beliefs: $q(s_{t+1} | o_{t+1})$ using Bayesian updating
3. Evaluate EFE with these updated future beliefs

This recursive structure means the agent anticipates how its own beliefs will change — a form of meta-cognition. The recursion can be extended to arbitrary depth: the agent models what it will believe, about what it will believe, about what it will believe...

3. Hierarchical POMDPs

For multi-timescale planning, the generative model is organized hierarchically:

Level 1 (fast): $p(o | s_1, a_1)$ — moment-to-moment actions (e.g., individual keystrokes) **Level 2 (slow):** $p(s_1 | s_2, a_2)$ — mid-level sub-goals (e.g., typing a word) **Level 3 (slowest):** $p(s_2 | s_3, a_3)$ — high-level goals (e.g., writing a paragraph)

Higher levels set contexts (prior preferences, initial states) for lower levels. The temporal scale of each level is determined by its time constant — higher levels evolve more slowly.

The free energy minimization operates across levels: $\mathbf{F}_{\text{total}} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$

with top-down and bottom-up messages connecting levels, analogous to the perceptual hierarchy but extended to the action-planning domain.

Derivation Exercises

1. Derive the EFE for a two-step, two-action POMDP. Enumerate all policies and compute G for each.
2. Show how sophisticated inference modifies the EFE by incorporating future belief updates into the evaluation.
3. Write the generative model for a two-level hierarchical POMDP and derive the free energy for each level.

Conclusion

Planning completes the Active Inference framework by extending inference into the future — from single-step EFE evaluation through sophisticated recursive belief simulation to multi-timescale hierarchical planning. Together, the eight modules provide the complete mathematical foundation: from Markov Blankets (systems) through generative models (agents), perception, precision, action, learning, communication, and planning — a unified mathematics of mind.